

4 (a) free: (body oscillates) without any loss of energy/no resistive forces/no external forces applied

forced: continuous energy input (required)/body is made to vibrate by an (external) periodic force/driving oscillator

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- (b) (i) idea of resonance B1
maximum amplitude at natural frequency B1
frequency = 2.1 Hz (*allow 2.08 to 2.12 Hz*) B1 [3]
- (ii) peak not very sharp/amplitude not infinite so frictional forces are present B1 [1]
- (c) $v = \omega x_0$
 $= 2\pi \times 2.1 \times 4.7 \times 10^{-2}$ (*allow ECF from (b)(i)*) C1
 $= 0.62 \text{ m s}^{-1}$ A1 [2]